

مسائل حل

## Circular Motion

4m wire at left has  $I = 2A$  in B. If  $F_B = 32N$  down. Find B

$$B = \frac{32}{4 \times 2} = 4 \text{ inward}$$

Metallic

سوال ۱۸۱

## Circular Motion

4m wire at right has  $I = 2A$  in B. If  $F_B = 32N$  up. Find B

$$B = \frac{32}{4 \times 2} = 4 \text{ (inward)}$$

Metallic

20/12/2019

## Circular Motion

\*  $F_B$  is maximum  $\rightarrow F_B = \frac{1}{2} v B$

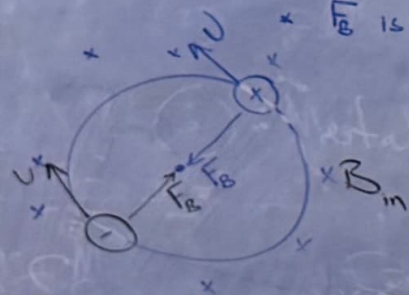
\*  $F_B$  is directed toward a center  $\rightarrow$  Centrifugal force

if  $B$  is inward:

① Positive: rotates counter clock wise

Negative: rotates clock wise

if  $B$  is outward:



$$f = \frac{\text{no. of cycles}}{\text{total time}}$$

$$T = \frac{\text{total time}}{\text{no. of cycles}}$$

$$T = \frac{1}{f}$$

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$$F_B = \frac{mv^2}{r}$$

$$F_B = \frac{1}{2} v B$$

$$\frac{1}{2} v B = \frac{mv^2}{r}$$

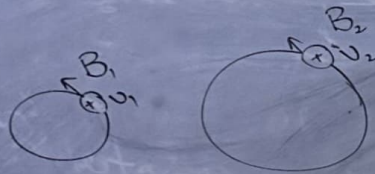
$$\frac{1}{2} B r = m v \rightarrow \frac{1}{2} B r = m \omega$$

$$\omega = \frac{v}{r}, \quad \omega = 2\pi f, \quad \omega = \frac{2\pi}{T}$$

$f \rightarrow$  frequency is the number of cycles per 1 sec.  
 $T \rightarrow$  Periodic time is the time to complete 1 cycle

سوال ۱۰

$$\oint \mathbf{B} \cdot \mathbf{r} = \mu_0 I$$

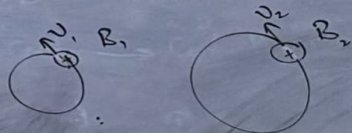


$$I_1 = I_2$$

$$\therefore r_2 > r_1$$

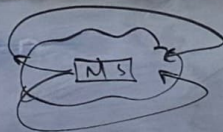
$$\therefore B_2 < B_1$$

$$B_1 = B_2$$



$$\therefore r_2 > r_1$$

$$\therefore I_2 > I_1$$



Gauss's law for magnetism:

\* Total magnetic flux through any closed surface is zero:

$$\oint \mathbf{B} \cdot \mathbf{d}\mathbf{A} = 0$$

\* mono magnetic pole does not exist

Gauss's law for electricity:

\* Total electric flux through any closed surface is not zero

$$\oint \mathbf{E} \cdot \mathbf{d}\mathbf{A} = \frac{Q_{\text{in}}}{\epsilon_0}$$

\* mono charge does exist

Ques

A proton is moving in circular orbit of radius  $r = 21 \text{ cm}$  in  $B = 0.42 \text{ T}$  perpendicular to  $v$ .  
The time to complete one cycle is:

$$\frac{q}{m} B = \omega$$

$$\omega = \frac{qB}{m}$$

$$T = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi m}{\frac{q}{m} B} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.42} = 1.56 \times 10^{-7} \text{ s}$$